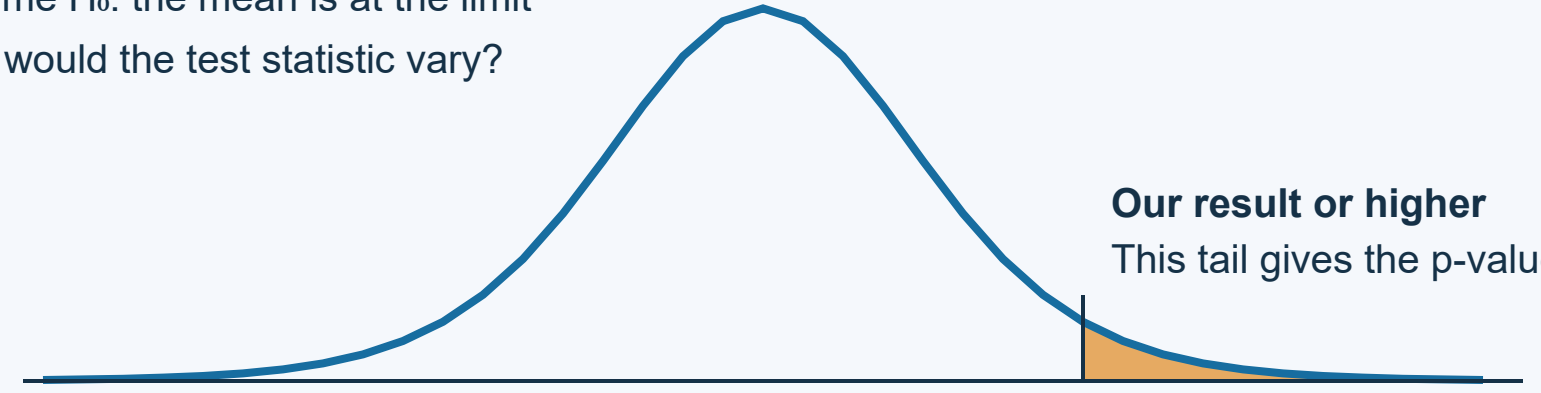
Assume H_0: the mean is at the limit

How would the test statistic vary?